CO1 – HOME ASSIGNMENT QUESTIONS

1. A dataset has the following descriptive statistics: Mean = 68, Median = 61, Standard Deviation = 14 , Pearson's coefficient of skewness: $Sk = \frac{3(\text{Mean}-\text{Median})}{\text{Standard Deviation}}$. Also, the kurtosis coefficient is 4.8.
   a) Calculate the coefficient of skewness.
   b) Determine the distribution as positively skewed, negatively skewed, or symmetric.
   c) Determine whether the distribution is platykurtic, mesokurtic, or leptokurtic

2. The following frequency distribution represents the ages of customers visiting a retail store.

Age (Years)	Frequency
20–30	4
30–40	8
40–50	15
50–60	10
60–70	6
70–80	2

   a) Draw a histogram for the given data.
   b) Determine whether the distribution is approximately symmetric, positively skewed, or negatively skewed.
   c) Determine whether the mean or median is likely to be larger and justify your answer.

3. The weights (kg) of 15 students are: 42, 45, 48, 50, 52, 55, 58, 60, 62, 65, 67, 70, 72, 75, 78. Calculate: Mean, Median, Standard Deviation. Interpret the results.
4. The monthly sales (units) are: 120, 125, 128, 130, 132, 135, 138, 140, 145, 150. Calculate: Variance and Standard Deviation. Comment on the consistency of sales.
5. The following summary statistics are obtained: Minimum = 15 , Q_1 = 25 , Median = 35 , Q_3 = 48 , Maximum = 75. Calculate: IQR, Lower Fence, Upper Fence. Draw the corresponding Box Plot and identify any outliers.
6. The annual income (₹ lakh) of employees is: 3.5, 4.0, 4.2, 4.5, 4.8, 5.0, 5.3, 5.5, 18.0, 22.0. Calculate: Mean, Median. Which measure better represents the data? Justify your answer.
7. The daily sales (₹ thousand) of a retail store are: 45, 48, 50, 52, 55, 58, 60, 62, 65, 120. Calculate: Mean, Median, Standard Deviation. Comment on the effect of the extreme value on the measures of central tendency.
8. The waiting times (minutes) of customers are: 2, 3, 4, 5, 6, 7, 8, 10, 12, 25. Calculate: Mean, Median, IQR. Comment on the skewness and identify any outliers.
9. Differentiate between a Histogram and a Box Plot.
10. Explain the basic steps involved in Exploratory Data Analysis (EDA).